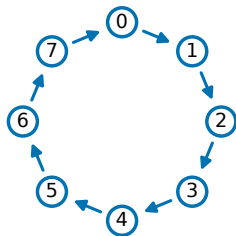


Transition families across one, two, and three latent dimensions

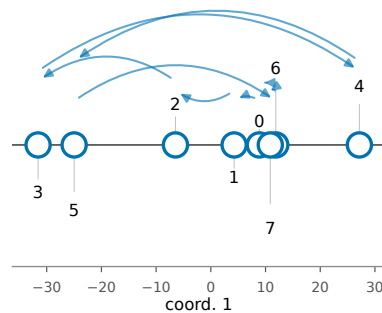
transition family

Cycle

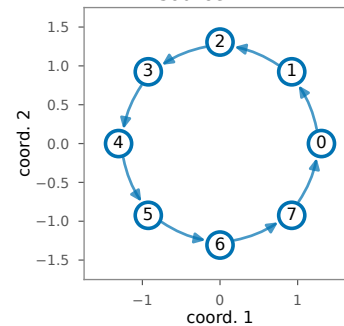


1 cyclic action

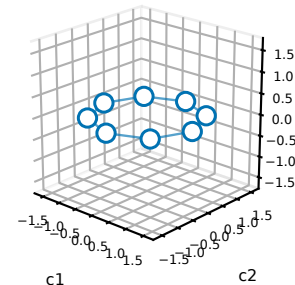
m=1
gain 2.35 | MSE 109.444
source m=1



m=2
gain 1.00 | MSE 0
source m=2



m=3
gain 1.00 | MSE 0
from m=2, extra coord.=0

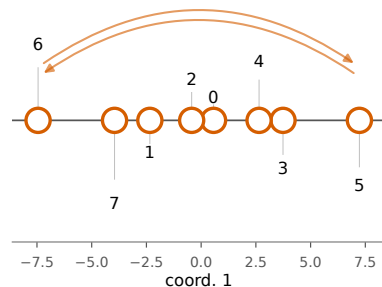


Adjacent pair actions

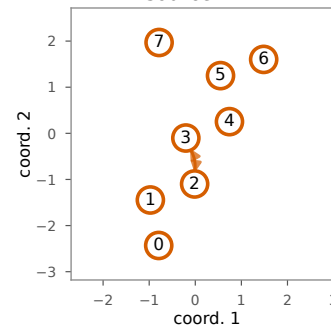


7 neighboring-pair actions

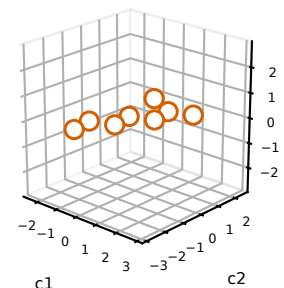
gain 3.21 | MSE 2.322
source m=1



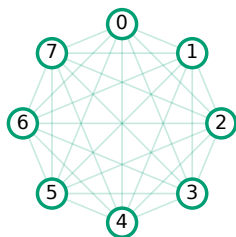
gain 1.56 | MSE 0.094
source m=2



gain 1.56 | MSE 0.094
from m=2, extra coord.=0

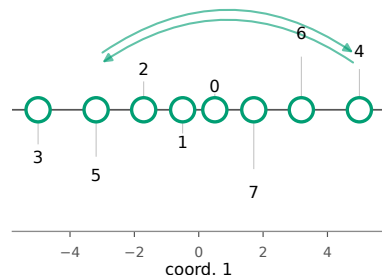


All-pair actions

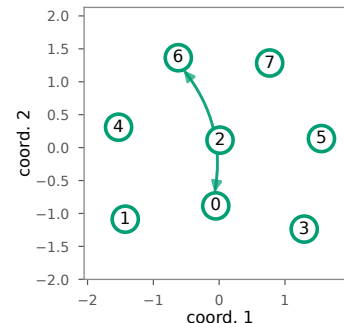


28 all-pair actions

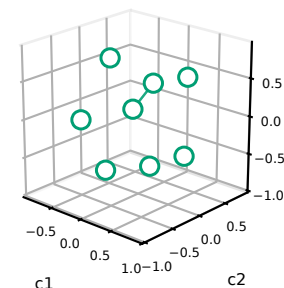
gain 5.54 | MSE 2.214
source m=1



gain 2.32 | MSE 0.269
source m=2



gain 1.56 | MSE 0.045
source m=3



L=1 MSE is selected from completed source dimensions $k \leq m$; a higher dimension may leave extra coordinates unused.